University of the Western Cape

Department of Mathematics and Applied Mathematics

MAT211 Advanced Calculus

Tutorial 8 **29 September 2026**

Vector Function and Space Curves

Question 1

Find the domain of the vector function

a) $\boldsymbol{r}(t) = \langle \sqrt{9-t^2}, e^{-3t}, \ln(t+3) \rangle$.

b) $\boldsymbol{r}(t) = \frac{t-2}{t+2}\hat{\mathbf{i}} + \cos t\, \hat{\mathbf{j}} + \ln(9-t^2)\hat{\mathbf{k}}$.

Question 2

Find a vector equation and parametric equation for the line segment that joins A to B if A (1,0,1) and B (2,3-3).

Question 3

Two particles travel along the space curves $\boldsymbol{r_1}(t) = \langle t\ ,\ t^2\ , t^3 \rangle$ and $\boldsymbol{r_2}(t) = \langle 1+2t\ , 1+6t\ , 1+14t \rangle$

Do the particles collide?

Do their paths intersect?

Question 4

Find a vector function that represents the curve of intersection of the two surfaces.

a) The cylinder $x^2 + y^2 = 9$ and the surface $z = xy$.
b) The paraboloid $z = 4x^2 + y^2$ and the parabolic cylinder $y = x^2$.

Derivatives and Integrals of Vector Functions

Question 1

If $\boldsymbol{r}(t) = \langle 1-t\ , \sqrt{t} \rangle$, find $\boldsymbol{r}'(t)$.

Sketch the position vector $\boldsymbol{r}(t)$ and the tangent vector $\boldsymbol{r}'(t)$ when $t = 1$.

Question 2

Find the derivatives of the vector functions

a) $\boldsymbol{r}(t) = \langle \tan(2t), \cos t\ , -t^2 \rangle$.

b) $\boldsymbol{r}(t) = \langle at\ \cos(2t)\ \mathbf{i} + b\sin^2 t\ \mathbf{j} + c\tan^2 t\ \mathbf{k} \rangle$.

Question 3

Find the unit tangent vector $\boldsymbol{T}(t)$ at $t = 1$ for $\boldsymbol{r}(t) = \langle te^{-t}, 2\tan t, 3\sin t\rangle$.

Question 4

Find the point on the curve $\boldsymbol{r}(t) = \langle 2\cos t, 2\sin t, e^{t}\rangle$, $0 \le t \le \pi$, where the tangent line is parallel to the plane $\sqrt{3}x + y = 1$.

Vector Fields and Gradient Fields

Question 1

Sketch the vector field $\boldsymbol{F}$ in each case

a) $\boldsymbol{F}(x, y) = 2y\,\hat{\mathbf{i}} + 3x\,\hat{\mathbf{j}}$

b) $\boldsymbol{F}(x, y) = \frac{y\hat{\mathbf{i}} - x\hat{\mathbf{j}}}{\sqrt{x^2+y^2}}$

Question 2

At time $t = 1$, a particle is located at position (1,3). If it moves in a velocity field $\boldsymbol{F}(x, y) = \langle xy - 2, y^2 - 10\rangle$, find its approximate location at time $t = 1.04$.

Question 3

Find the gradient vector field ∇f of f in each case:

a) $f(x, y) = x^3 - 2xy$.

b) $f(x, y) = (3x^2 + y^3)^{\frac{1}{2}}$.

c) $f(x, y, z) = x^2 \ln(y - 4z)$.

Line Integrals

Question 1

Evaluate the line integrals where C is the given curve.

a) $\int_C xy\,\mathrm{ds}$ where C is defined by $x = t^2$, $y = 2t$ where $0 \le t \le 1$.

b) $\int_C x\,\sin y\,\mathrm{ds}$ where C is the line segment from (0,3) to (4,6).

c) $\int_C \sin 2x\,\mathrm{d}x + \cos 2y\,\mathrm{dy}$ where C consists of the top half of the circle $x^2 + y^2 = 1$ from (1,0) to (-1,0) and the line segment from (-1,0) to (-2,3).

Question 2

Evaluate the line integral $\int_C \boldsymbol{F}.d\boldsymbol{r}$ where C is given by the vector function $\boldsymbol{r}(t)$.

a) $\boldsymbol{F}(x,y,z) = (x+2y)\,\hat{\mathbf{i}} + (y-z)\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$ where $\boldsymbol{r}(t) = t^2\,\hat{\mathbf{i}} + t^3\,\hat{\mathbf{j}} + t^2\,\hat{\mathbf{k}}$ for $0 \le t \le 1$.

b) $\boldsymbol{F}(x,y,z) = z\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} - 2x\,\hat{\mathbf{k}}$ with $\boldsymbol{r}(t) = 2t\,\hat{\mathbf{i}} + \cos t\,\hat{\mathbf{j}} + \sin t\,\hat{\mathbf{k}}$ for $0 \le t \le \pi$.

Fundamental Theorem for Line Integrals

Question 1

Determine whether or not $\boldsymbol{F}$ is a conservative vector field. If it is, find a function f such that $\boldsymbol{F} = \nabla f$.

a) $\boldsymbol{F}(x,y) = e^x \cos y\,\hat{\mathbf{i}} - e^x \sin y\,\hat{\mathbf{j}}$.

b) $\boldsymbol{F}(x,y) = (3x^2 - y^2)\,\hat{\mathbf{i}} + (4xy - 2)\,\hat{\mathbf{j}}$.

c) $\boldsymbol{F}(x,y) = (ye^x + \sin y)\,\hat{\mathbf{i}} + (e^x + x\cos y)\,\hat{\mathbf{j}}$.

Question 2

Find a function f such that

a) $\boldsymbol{F} = \nabla f$

b) Use (a) to evaluate $\int_C F.\,\mathrm{d}\boldsymbol{r}$ along the given curve C if $\boldsymbol{F}(x,y) = x^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}}$ and C is the arc of the parabola $y = 2x^2$ from (-1,2) to (2,8).

Question 3

Show that the line integral is independent of path and evaluate the integral for $\int_C \tan y\,\mathrm{dx} + x\sec^2 y\,\mathrm{dy}$, C is any path from (1,0) to $\left(2, \frac{\pi}{4}\right)$.

Question 4

Given $\boldsymbol{F}(x,y) = \frac{y\,\hat{\mathbf{i}} + x\hat{\mathbf{j}}}{x^2+y^2}$, find if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

Green's Theorem

Question 1

Evaluate the line integral $\oint_C xy\,\mathrm{d}x + x^2\,\mathrm{d}y$ using two methods:

a) directly.
b) using Green's Theorem.

C is the rectangle with vertices (0,0), (3,0), (3,1) and (0,1).

Question 2

Use Green's Theorem to evaluate $\int_C F.\,\mathrm{d}\boldsymbol{r}$ for the following cases:

a) $\boldsymbol{F}(x,y) = \langle y^2\cos x\,, x^2 + 2y\sin x\rangle$, C is the triangle from (0,0) to (2,6) to (2,0) to (0,0).
b) $\boldsymbol{F}(x,y) = \langle e^x + x^2 y\,,\ e^y - xy^2\rangle$, C is the circle $x^2 + y^2 = 25$ oriented clockwise.